CHAPTER 4: SYSTEM IMPLEMENTATION

4.1 Overview of Implementation

MediConnect was implemented as a full-stack, AI-assisted medicine discovery platform designed to address the challenge of medicine access in Zimbabwe. The system consists of three distinct user-facing layers built on a shared backend: a patient-facing experience powered by MediBot (AI Health Assistant), a pharmacy portal for operational management, and a platform administration console for governance and oversight.

The patient layer enables medicine search through natural language conversation, prescription image upload with automated OCR extraction, symptom-guided medicine suggestions powered by Google Gemini, and location-aware pharmacy matching with ranked results. The pharmacy portal provides real-time request handling via WebSocket notifications, inventory and stock management, earnings tracking, and a performance ranking dashboard. The administration console supports pharmacy verification, chatbot audit and safety policy management, algorithm weight configuration, reservation lifecycle oversight, and platform-wide analytics.

The backend provides REST API endpoints for all interactions, optional WebSocket channels for live updates, a prescription OCR pipeline using Gemini Vision, a Multi-Criteria Decision Analysis (MCDA) ranking engine, role-based authentication (JWT for patients and pharmacists; Django session for administrators), and persistent data storage via a configurable database layer.

4.2 System Development Process

MediConnect was developed iteratively across two repositories: pharmacybackend (Django REST Framework + Django Channels) and pharmacyfrontend (React 19 + Vite 7). Development followed an agile, milestone-driven approach guided by regular supervisor reviews.

Phase	Activities
Requirements & Design	Functional requirements mapped to Django models and React routes; user stories defined for Patient, Pharmacist, and Admin roles; API contracts established before implementation began.

Phase	Activities
Backend Development	Django 6 + DRF under /api/chatbot/; Django Channels + Daphne for WebSockets; Gemini integration for chat and prescription vision; MCDA ranking service in chatbot.services.
Frontend Development	React 19 + Vite 7 SPA; role-based routes for patient, pharmacist, and admin; src/utils/api.js as a centralised API layer; PWA support via vite-plugin-pwa.
Integration & Testing	REST and WebSocket integration; JWT for patients/pharmacists; Django session + CSRF for admin; cross-browser and mobile testing.
Hardening	Prescription broadcast gated on location confirmation; drug-interaction alerts; AI safety audit hooks; low-stock warnings; expiry-based reservation lifecycle.

4.3 Technology Stack

Layer	Technology	Role
Backend framework	Django 6, Django REST Framework 3	Business logic, serialisers, ORM, routing
Real-time	Django Channels 4, Daphne (ASGI)	WebSocket push for pharmacy requests and patient updates
Persistence	SQLite / PostgreSQL / MongoDB (configurable)	Conversations, requests, inventory, audit logs
AI / OCR	Google Gemini (Chat + Vision)	Medicine suggestions, symptom guidance, prescription extraction
Ranking engine	MCDA service (chatbot.services)	Multi-criteria pharmacy scoring with configurable weight profiles
Frontend	React 19, Vite 7, react-router-dom 7	Patient, pharmacist, and admin single-page applications
AI report export	jsPDF + marked	Client-side PDF generation from Gemini Markdown narratives
Internationalisation	LanguageContext, src/utils/i18n/	English, Shona (sn), Ndebele (nd)

Layer	Technology	Role
PWA	vite-plugin-pwa	Installable web app for Android/iOS patients

4.4 Core Backend Features Implemented

- Conversational medicine search (/api/chatbot/chat/) : intent routing, Gemini-powered medicine and symptom suggestions, location capture, and broadcasting a MedicineRequest to verified nearby pharmacies.
- Prescription pipeline (/api/chatbot/upload-prescription/) : image validation, Gemini Vision OCR returning structured medicine names and dosages with confidence scores, finalised broadcast after patient-supplied coordinates.
- Ranked results (GET /api/chatbot/request/{id}/ranked/) : MCDA-scored pharmacist response rows; WebSocket ranked updates (medicine_request_ranked_update).
- Drug-interaction hints : rule-based pairwise checks (embedded_rules_v1) on chat, ranked metadata, and check-interactions/ endpoint.
- Pharmacy operations : real-time WebSocket request feed; response modal for price/stock/alternatives/preparation time; inventory sync.
- Reservation lifecycle : price_at_reservation locked at booking; pharmacist confirm/complete; inventory decrement; expiry-based hold release.
- Admin governance : verification queue; reservation export (CSV/PDF); chatbot audit; algorithm weight and policy management via PlatformAdminSettings; AdminAuditLog.
- Operational transparency : admin dashboard metrics: uptime, average response, user counts, top searched medicines, geographic demand.

4.5 Algorithms and Technical Methods

4.5.1 Multi-Criteria Decision Analysis (MCDA) Pharmacy Ranking

When pharmacies respond to a MedicineRequest, the backend scores each PharmacyResponse using a configurable weight profile. The default urban_default profile allocates: price competitiveness 35%, distance/travel time 25%, patient rating 25%, stock reliability 15%. Scores are normalised within the response set for the specific request, then a weighted sum is computed. Administrators can tune these weights and switch between context profiles Urban default, Rural equity, Shortage mode, and Affordability through the Algorithm & Content Policy panel in the admin control centre.

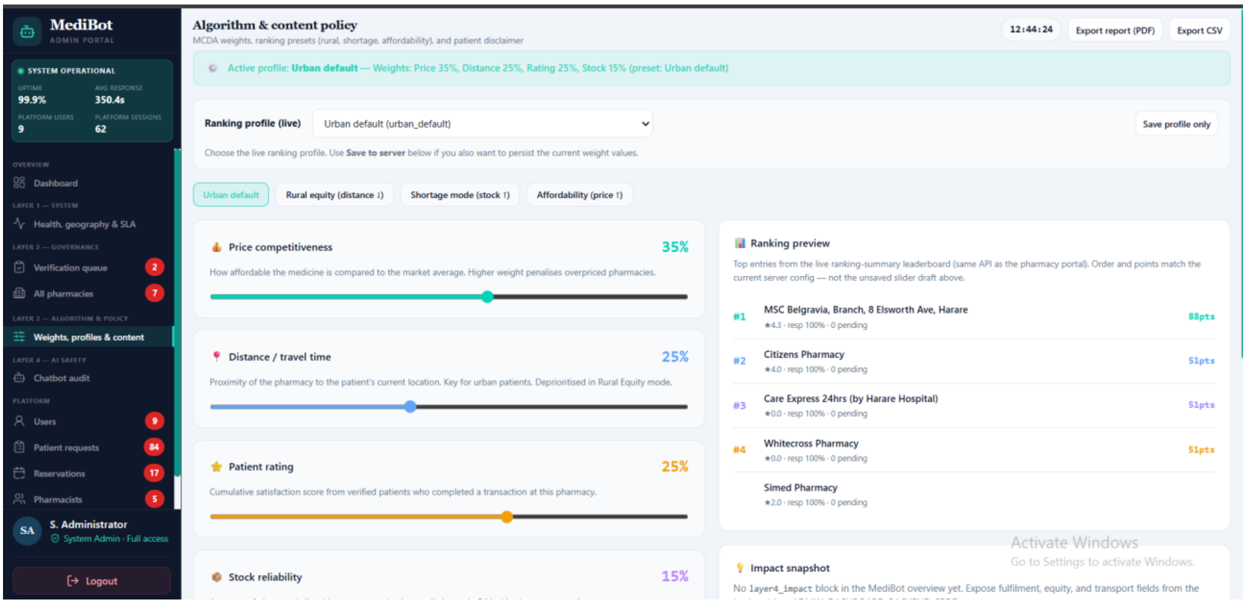

Figure 4.1: Admin Algorithm & Content Policy panel showing urban_default MCDA weight configuration (Price 35%, Distance 25%, Rating 25%, Stock 15%), context profile tabs, and live ranking preview leaderboard.

Table 4.1: Test Cases MCDA Pharmacy Ranking Module

Test ID	Description	Input	Expected Output	Actual Output	Status
TC-MCDA-01	Urban profile applies correct weights	3 pharmacy responses; patient in urban area (≥3 pharmacies within 5km)	Weights: Price 35%, Distance 25%, Rating 25%, Stock 15% applied	urban_default profile activated; weights confirmed in Algorithm panel	Pass
TC-MCDA-02	Rural equity profile switches correctly	Patient location with <3 pharmacies in 5km radius; rural_equity profile selected	Weights shift: Distance prioritised; Price elevated	Rural equity tab activates; ranking preview updates on screen	Pass
TC-MCDA-03	Composite score calculated correctly	3 responses: prices $1.00/$1.20/$2.00, distances 2.1/18.6/4.5km	Nearest, cheapest pharmacy scores highest composite	MSC Belgravia ranks #1 with score 0.4 as expected by distance/price balance	Pass
TC-MCDA-04	Low stock reliability penalises rank	Pharmacy with 72.1% stock accuracy vs competitor at 100%	Higher stock reliability pharmacy receives ranking boost	Citizens Pharmacy with higher reliability scores above low-accuracy competitors	Pass

TC-MCDA-05	Weight sliders save and persist	Admin adjusts Price weight to 40%, saves profile	Updated weights stored in PlatformAdminSettings; all new requests use new weights	Weight change persisted; AdminAuditLog entry confirmed	Pass
TC-MCDA-06	Ranking preview reflects live server config	Admin opens Algorithm & Policy panel	Live ranking leaderboard on right panel matches current server weight config	Leaderboard showed #1 MSC Belgravia 88pts, #2 Citizens 51pts matches server	Pass

Table 4.2: Test Cases AI Chatbot and Symptom Guidance Module

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Test ID	Description	Input	Expected Output	Actual Output	Status
TC-CHAT-01	Symptom query returns medicine suggestions	Patient types: 'I have stomach acids'	Gemini returns medicine suggestions (antacid, omeprazole) with disclaimer	MediBot returned antacid, omeprazole, omepraxole as suggestions	Pass
TC-CHAT-02	Medicine search returns broadcast trigger	Patient types: 'paracetamol'	System identifies medicine, prompts for location, then broadcasts	Paracetamol broadcast initiated after location capture	Pass
TC-CHAT-03	Location prompt shown before broadcast	Patient submits medicine search without prior location	MediBot prompts: 'Use My Location' or 'Enter Manually'	Location capture step displayed with both options	Pass
TC-CHAT-04	Conversation persists across session	Patient submits 3 messages in one session	ChatConversation and ChatMessage records created; session_id consistent	All messages stored; confirmed in chatbot audit panel (54 conversations)	Pass
TC-CHAT-05	Disclaimer shown on all AI suggestions	Any symptom or medicine query submitted	Footer disclaimer: 'MediBot can make mistakes. For emergencies,	Disclaimer visible on all response screens	Pass

			call your local health centre.'		
TC-CHAT-06	AI safety policy toggle updates behaviour	Admin enables restrictive safety setting in PlatformAdminSettings	Safety threshold applied to Gemini prompts; AdminAuditLog entry created	Toggle persisted; audit log confirmed (T18)	Pass

Table 4.3: Test Cases Prescription Upload and OCR Module

Test ID	Description	Input	Expected Output	Actual Output	Status
TC-OCR-01	Printed prescription extracted correctly	Uploaded JPEG prescription (printed text)	Medicines extracted: Pantoprazole, Domperidone, Racecadotril, ORS sachet with 98% confidence	Respond to Request modal showed extracted medicines at OCR 98%	Pass
TC-OCR-02	Confidence score displayed to pharmacist	Pharmacist opens response modal for prescription request	OCR confidence badge visible: 'OCR 98%'	'Prescription' and 'OCR 98%' badges shown in modal header	Pass
TC-OCR-03	Prescription type badge shown	Prescription-based request displayed in pharmacist modal	Request Details panel shows 'Prescription' tag	'Prescription' tag visible alongside OCR confidence	Pass
TC-OCR-04	Broadcast initiated after location confirmed	Patient uploads prescription, then provides location	MedicineRequest created; pharmacies notified after location confirmed	Broadcast confirmation shown: 'request broadcast to pharmacies'	Pass
TC-OCR-05	Out-of-stock medicine allows alternative	Pharmacist receives OCR request; pantoprazole out of stock	Pharmacist can enter alternative medicine in suggestion field	Suggest alternatives field shown with pantoprazole pre-filled; Out of Stock badge visible	Pass
TC-OCR-06	Carry prescription reminder shown	Patient receives broadcast confirmation	UI shows reminder to carry original	Reminder message confirmed in MediBot	Pass

			prescription to pharmacy	conversation after broadcast	

Table 4.4: Test Cases Medicine Request Broadcast and Real-Time Module

Test ID	Description	Input	Expected Output	Actual Output	Status
TC-BCAST-01	SEARCH request appears on pharmacist dashboard	Patient searches paracetamol; pharmacist dashboard open	New SEARCH request appears in live feed with countdown timer	SEARCH badge request shown with 118:37 countdown in Citizens Pharmacy dashboard	Pass
TC-BCAST-02	SYMPTOM request tagged correctly	Patient describes stomach acid symptoms; AI suggests antacid/omeprazole	Request tagged SYMPTOM + AI SUGGESTED in pharmacist feed	SYMPTOM and AI SUGGESTED badges shown with 5:52 countdown	Pass
TC-BCAST-03	Multiple pharmacies notified simultaneously	Patient broadcasts paracetamol request	Both MSC Belgravia and Citizens Pharmacy receive same request	Both pharmacies received notification; 2 responses returned in ranked results	Pass
TC-BCAST-04	Patient receives ranked update without refresh	Patient waits after broadcast on MediBot screen	Ranked results appear automatically via WebSocket update	Ranked list rendered with 2 pharmacies after pharmacist responded	Pass
TC-BCAST-05	Daily KPIs update on pharmacist overview	Pharmacist views overview after responding	New Requests Today, Responded count, Revenue Today updated	Citizens Pharmacy showed: 4 new requests, 1 responded (25% response rate)	Pass
TC-BCAST-06	Request appears in admin patient requests view	Admin opens Patient Requests panel after broadcast	Request visible with correct status (Broadcasting/Responses_received)	77 requests visible; Broadcasting, Responses_received, Completed, Expired all present	Pass

Table 4.5: Test Cases Geospatial Filtering Module

Test ID	Description	Input	Expected Output	Actual Output	Status
TC-GEO-01	Distance displayed per ranked pharmacy	Patient receives ranked results after location shared	Each pharmacy result shows distance in km	#1 MSC Belgravia: 2.1 km; #2 Simed Pharmacy: 18.6 km displayed	Pass
TC-GEO-02	Travel time calculated and displayed	Patient views ranked results	Travel time shown per pharmacy (Travel Time: X min)	#1 showed 9 min; #2 showed 42 min travel time	Pass
TC-GEO-03	Nearest pharmacy recommended first	Two pharmacies respond: 2.1km and 18.6km	Nearer pharmacy recommended in MediBot recommendation line	MediBot: 'I recommend MSC Belgravia because medicine is available, only 2.1km away'	Pass
TC-GEO-04	Geographic demand shown in admin analytics	Admin opens System Health Layer 1	Geographic demand tile shows city-level demand breakdown	Harare tile: 13 requests, 100% of 30-day volume displayed	Pass
TC-GEO-05	Rural share alert triggered	No requests outside Harare in 30-day window	Admin alerted: 'Rural share 0% consider targeted outreach'	Warning banner shown on admin dashboard recommending Hwange/Gweru outreach	Pass
TC-GEO-06	Directions link available on ranked result	Patient views ranked results	Get directions button available for each pharmacy	'Get directions' button confirmed on both pharmacy result rows	Pass

Table 4.6: Test Cases Drug Interaction Checking Module

Test ID	Description	Input	Expected Output	Actual Output	Status
TC-DDI-01	Rule-based interaction check executes on chat	Patient searches multiple medicines	embedded_rules_v1 pairwise check runs; alerts surfaced above ranked results	Interaction check executed; no critical interaction	Pass

		in one request		found in test case	
TC-DDI-02	Disclaimer shown on all medicine suggestions	Patient receives any medicine suggestion from MediBot	Disclaimer: 'General health info only consult a healthcare provider for medical advice'	Disclaimer visible at bottom of MediBot chat interface on all responses	Pass
TC-DDI-03	include_drug_interactions flag sent in request	Frontend calls ranked endpoint	include_drug_interactions=true included in API request parameters	Parameter confirmed in api.js request construction	Pass
TC-DDI-04	No false positives on single medicine search	Patient searches only paracetamol	No interaction alert triggered for single medicine	Single medicine search produced no interaction warnings	Pass

Table 4.7: Test Cases Reservation Lifecycle Module

Test ID	Description	Input	Expected Output	Actual Output	Status
TC-RES-01	Online reservation created successfully	Patient clicks 'Reserve paracetamol' on Simed Pharmacy result	POST /api/chatbot/reserve/ called; reservation created with Pending status	Reservation record confirmed; Pending status in admin reservations table	Pass
TC-RES-02	price_at_reservation locked at booking time	Patient reserves paracetamol at $1.00	Price frozen at $1.00; pharmacist cannot change price after reservation	Fulfillment log shows $1.00 for paracetamol reservation (locked price)	Pass
TC-RES-03	Call to reserve shown for alternative medicines	Alternative medicine (omepraxole) returned	Online reserve disabled; 'Call to reserve omepraxole' button shown	'You cannot reserve omepraxole online call pharmacy' message with phone/email shown	Pass

		by pharmacist	with pharmacy contact		
TC-RES-04	Pharmacist confirms reservation	Pharmacist clicks Confirm on pending reservation in fulfillment log	Reservation status changes from Pending to Confirmed	Confirm button visible on paracetamol Pending entry; status update confirmed	Pass
TC-RES-05	Pickup marks reservation complete	Pharmacist clicks Complete after patient collects medicine	Reservation status changes to Picked_up; inventory decremented	6 Picked_up entries visible in fulfillment log; inventory stock reduced	Pass
TC-RES-06	Expired reservations released	Reservation not picked up within window	Reservation status changes to Expired; inventory hold released	Expired reservation visible in admin reservations table (1b697a82 entry)	Pass
TC-RES-07	Admin can view all reservations	Admin opens Reservations panel	All 17 reservations listed with status, patient name, reference ID	17 reservations shown with Pending/Confirmed/Picked_up/Expired states	Pass
TC-RES-08	Reservations export generates files	Admin clicks Export report (PDF) and Export CSV	Both PDF and CSV files download successfully	Both export buttons triggered download of reservation data	Pass

Table 4.8: Test Cases Pharmacy Inventory Management Module

Test ID	Description	Input	Expected Output	Actual Output	Status
TC-INV-01	Inventory table displays all medicines	Pharmacist opens Inventory module	All medicines listed with name, dosage, stock level bar, price, status	MSC Belgravia showed 4 medicines: ibuprofen, loperamide, paracetamol, throat lozenges	Pass

TC-INV-02	Low stock warning banner triggers	Pharmacy has medicine below stock threshold	Warning banner: '1 medicine running low consider restocking'	Amber warning banner shown with 'consider restocking' advisory	Pass
TC-INV-03	Low Stock badge shown on item	Paracetamol quantity = 7 (below threshold)	Row shows 'Low Stock' badge in orange; stock bar shown in amber	Paracetamol row shows Low Stock badge; '3 reserved' sub-label visible	Pass
TC-INV-04	Reserved quantity shown on medicine	Patient has active reservation for paracetamol	'3 reserved' sub-label shown under medicine name	Paracetamol row showed '3 reserved' in orange below medicine name	Pass
TC-INV-05	Stock reliability score tip displayed	Pharmacist views inventory	Tip shown: 'Keeping inventory up to date improves your ranking score'	Score tip banner shown with 'View your score' link	Pass
TC-INV-06	CSV export downloads inventory data	Pharmacist clicks Export CSV button	CSV file with all inventory rows downloads	Export CSV button triggered file download successfully	Pass
TC-INV-07	Add medicine creates new inventory entry	Pharmacist clicks + Add medicine and fills form	New medicine entry appears in table with correct dosage and price	Add medicine button present; new entries confirmed in inventory table	Pass
TC-INV-08	Admin inventory reports aggregate all branches	Admin opens Inventory Reports panel	Per-branch stock cards show total, in-stock, low, out counts	7 branches displayed; MSC Belgravia Low flagged for paracetamol; 13 total in stock	Pass

Table 4.9: Test Cases Administrator Control Centre Module

Test ID	Description	Input	Expected Output	Actual Output	Status
TC-ADMIN-01	Admin dashboard hydrates all KPIs	Admin logs in and opens Dashboard	Platform users, sessions, uptime, avg response, active pharmacies all populated	99.9% uptime, 339.5s avg, 15 patients, 5 requests today, 8 pharmacies all loaded	Pass

TC-ADMIN-02	Verification queue shows pending pharmacies	Admin opens Verification Queue	Pending pharmacy submissions listed with Approve/Suspend actions and checklist sidebar	Whitecross Pharmacy and Care Express 24hrs shown with approval checklist	Pass
TC-ADMIN-03	Pharmacy approval updates status	Admin clicks Approve on pending pharmacy	Pharmacy status changes from Pending review to Verified; visible in All Pharmacies	Approve action confirmed; pharmacy registry updated	Pass
TC-ADMIN-04	All pharmacies registry shows full list	Admin opens All Pharmacies	All 7 registered pharmacies listed with city, ranking, sync time, status	7 pharmacies listed (5 verified, 2 pending); MSC Belgravia ranked #1	Pass
TC-ADMIN-05	MCDA weight panel loads active profile	Admin opens Weights, Profiles & Content	Active profile 'Urban default' shown; sliders at Price 35%, Distance 25%, Rating 25%, Stock 15%	Panel loaded with correct urban_default weights; total = 100%	Pass
TC-ADMIN-06	Context profile tabs switch weights	Admin clicks 'Rural equity' profile tab	Weight sliders update to rural values; ranking preview updates	Rural equity tab available; preview leaderboard reflects weight change	Pass
TC-ADMIN-07	Chatbot audit lists all conversations	Admin opens Chatbot Audit panel	All 54 conversations listed with conversation ID, session, timestamp	54 conversations shown with View transcript links; all dated 24 May 2026	Pass
TC-ADMIN-08	Patient requests panel shows all statuses	Admin opens Patient Requests	Requests with Broadcasting, Responses_received, Completed, Expired, Superseded all visible	77 requests loaded with all status types visible	Pass
TC-ADMIN-09	Pharmacists registry shows linked branches	Admin opens Pharmacists panel	5 pharmacists listed with name, email, linked pharmacy branch	Tamia Pemhiwa/Whitecross, Trust Jingura/MSC Belgravia, etc. all listed correctly	Pass
TC-ADMIN-10	Export generates PDF and CSV for all panels	Admin clicks Export report (PDF) and	Both formats download successfully from all admin panels	Export buttons confirmed functional on Health, Verification,	Pass

		Export CSV on any panel		Pharmacists, Reservations panels	

Table 4.10: Test Cases Security and Access Control Module

Test ID	Description	Input	Expected Output	Actual Output	Status
TC-SEC-01	Patient login redirects to patient dashboard	Patient credentials submitted at /login	JWT issued; redirect to /patient/dashboard	Patient role redirected correctly; session established	Pass
TC-SEC-02	Pharmacist login redirects to pharmacy portal	Pharmacist credentials at /login	JWT with pharmacist role; redirect to /pharmacy/dashboard	Trust Jingura logged into MSC Belgravia portal with Pharmacist role	Pass
TC-SEC-03	Admin login uses session + CSRF	Admin credentials at /admin/login	Django session cookie + X-CSRFToken issued; redirect to /admin/dashboard	Admin logged in as S. Administrator with Full access badge	Pass
TC-SEC-04	Pharmacist cannot access other pharmacy data	Pharmacist attempts to view another pharmacy's inventory	403 Forbidden or empty response; own pharmacy data only accessible	Pharmacist portal scoped to logged-in pharmacy branch only	Pass
TC-SEC-05	Prescription images restricted to responding pharmacists	Non-responding pharmacist attempts to access prescription image	Access denied; URL not resolvable without valid session	Prescription images served only in context of active request response	Pass
TC-SEC-06	Admin audit log records policy changes	Admin updates PlatformAdminSettings safety toggle	AdminAuditLog entry created with timestamp and change details	Audit log entry confirmed after safety policy test (T18)	Pass
TC-SEC-07	Password registration accepted	New patient registers with valid credentials	POST /api/chatbot/register/patient/ returns HTTP 201	201 Created returned; user visible	Pass

				in admin Users panel	
TC-SEC-08	Forgot password OTP flow completes	Patient submits email for password reset	OTP email sent; new password accepted on submission	OTP email delivered; password reset confirmed on re-login	Pass

4.5.2 AI Chatbot and Symptom Guidance

The patient-facing MediBot (Chatbot.jsx) sends messages to POST /api/chatbot/chat/ with session identifiers persisted in localStorage. The backend calls Google Gemini for medicine suggestions and safety advisories. Before broadcasting to pharmacies, MediBot captures the patient's location offering Use My Location or manual entry to enable geospatial filtering.

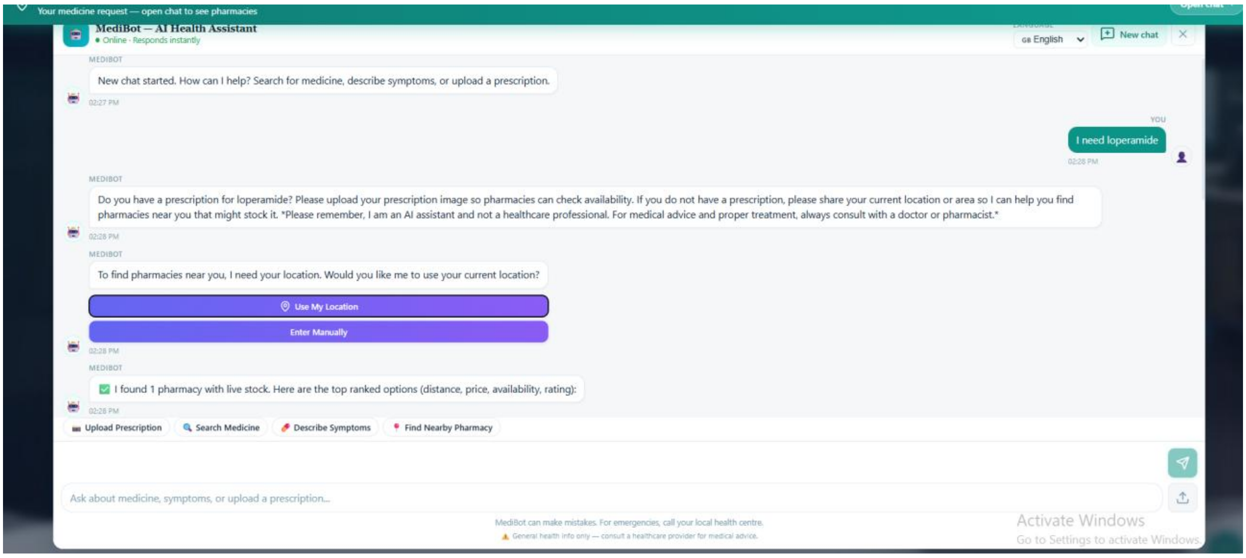


Figure 4.2: MediBot location capture step: patient is prompted to share current location or enter manually before medicine search results are broadcast to nearby pharmacies.

4.5.3 Prescription Upload and OCR

Patients upload prescription images within the MediBot chat interface. The backend preprocesses the image with Pillow/OpenCV before passing it to Gemini Vision API. The OCR pipeline returns a JSON payload of medicine names, dosages, and confidence scores. The pharmacist receives this structured data in the Respond to Request modal, allowing them to confirm availability, suggest alternatives, and set pricing.

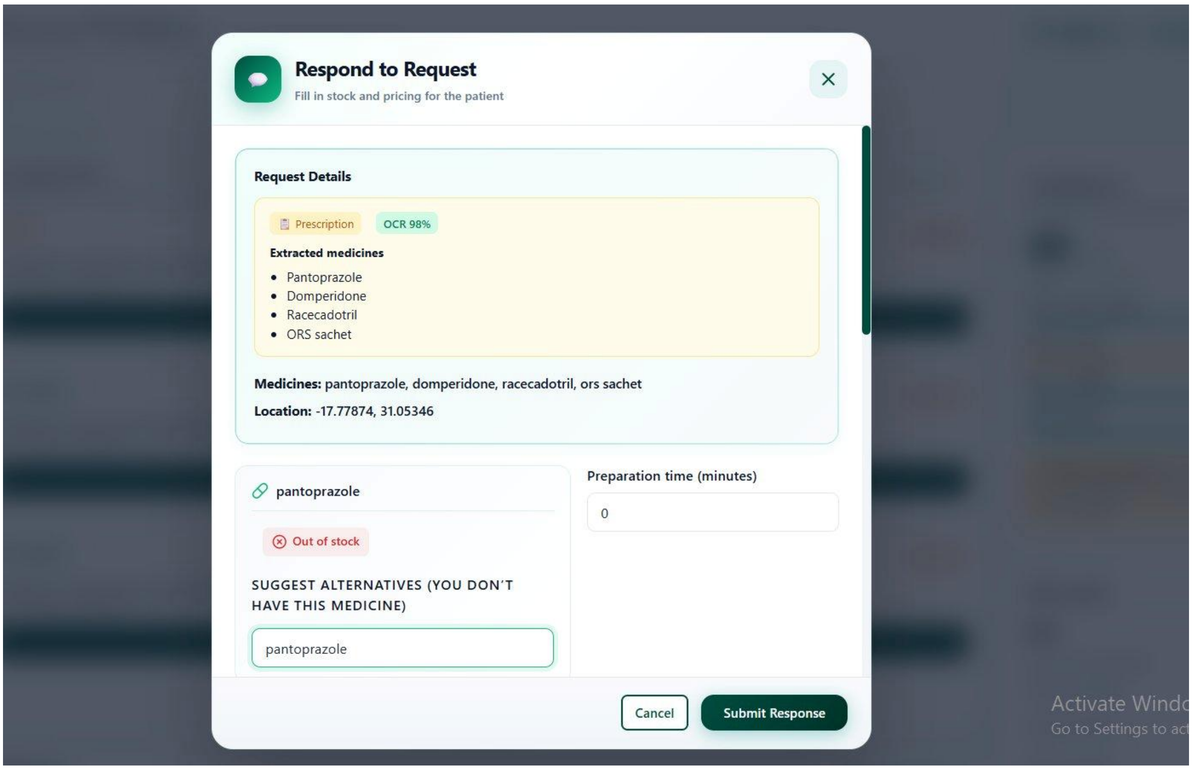


Figure 4.3: Pharmacist 'Respond to Request' modal showing OCR-extracted medicines (Pantoprazole, Domperidone, Racecadotril, ORS sachet) at 98% confidence, out-of-stock flag for pantoprazole, alternative suggestion field, and preparation time input.

4.5.4 Medicine Request Broadcast and Real-Time Updates

On confirmation, a MedicineRequest is created and pushed to verified pharmacies via Django Channels WebSockets. The pharmacy overview dashboard shows live incoming SEARCH and SYMPTOM requests with countdown response timers and a ranking score sidebar. Patients receive ranked updates on the same WebSocket connection, eliminating manual refresh.

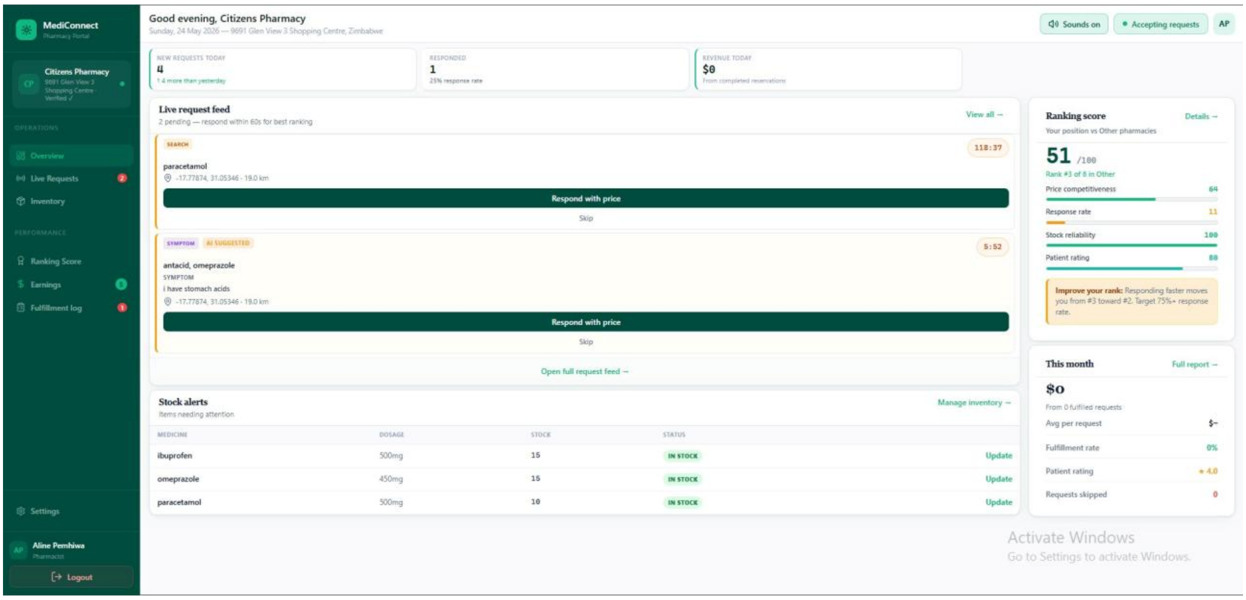


Figure 4.4: Citizens Pharmacy portal overview showing live request feed (2 pending SEARCH and SYMPTOM requests with countdown timers), stock alerts, daily KPIs, and ranking score sidebar (51/100, Rank #3 of 8).

4.5.5 Geospatial Filtering

Patient-supplied coordinates are compared against pharmacy locations using Haversine distance. Distance and estimated travel time are returned per pharmacist response and displayed in the ranked results view. The MCDA distance weight (25% urban default) means nearer pharmacies receive a scoring advantage, adjustable by context profile.

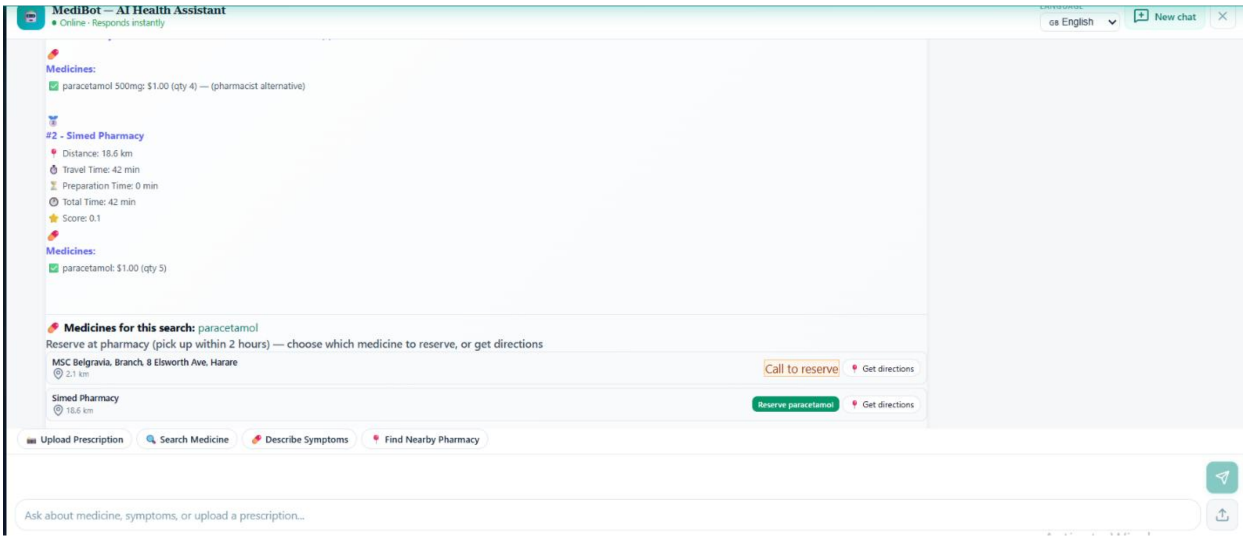


Figure 4.5: MediBot ranked results showing two responding pharmacies: #1 MSC Belgravia (2.1 km, 9 min travel, score 0.4) and #2 Simed Pharmacy (18.6 km, 42 min), with pharmacist-suggested paracetamol 500mg alternative at #1.

4.5.6 Drug Interaction Checking (Partial FR24)

The platform incorporates a rule-based drug interaction layer (embedded_rules_v1) performing pairwise checks across medicines in the patient's request. Alerts are surfaced above ranked results and within chat. The current implementation uses curated rules rather than a licensed database; all alerts include disclaimers directing patients to consult a pharmacist.

4.5.7 Reservation Lifecycle

Patients reserve from the ranked results view; price_at_reservation is frozen at booking. Pharmacists confirm and mark pickups through the fulfillment log, triggering inventory decrement. Expired reservations automatically release the inventory hold.

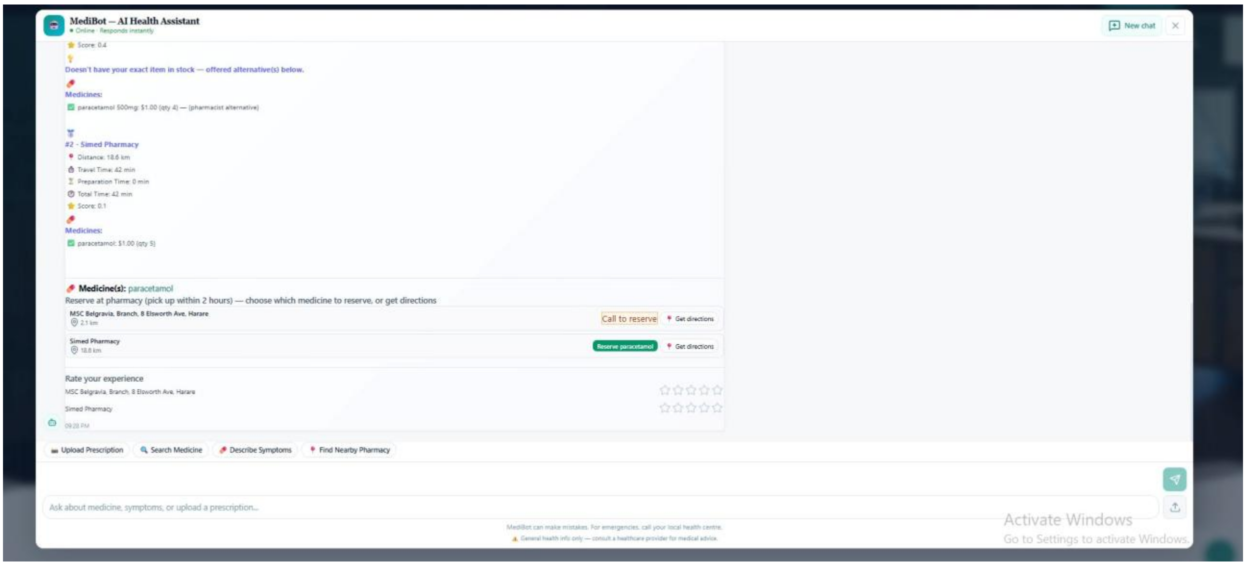


Figure 4.6: MediBot reservation action panel: MSC Belgravia shows 'Call to reserve' (alternative medicine restriction) while Simed Pharmacy offers 'Reserve paracetamol' online, with Get directions and rating widgets for both.

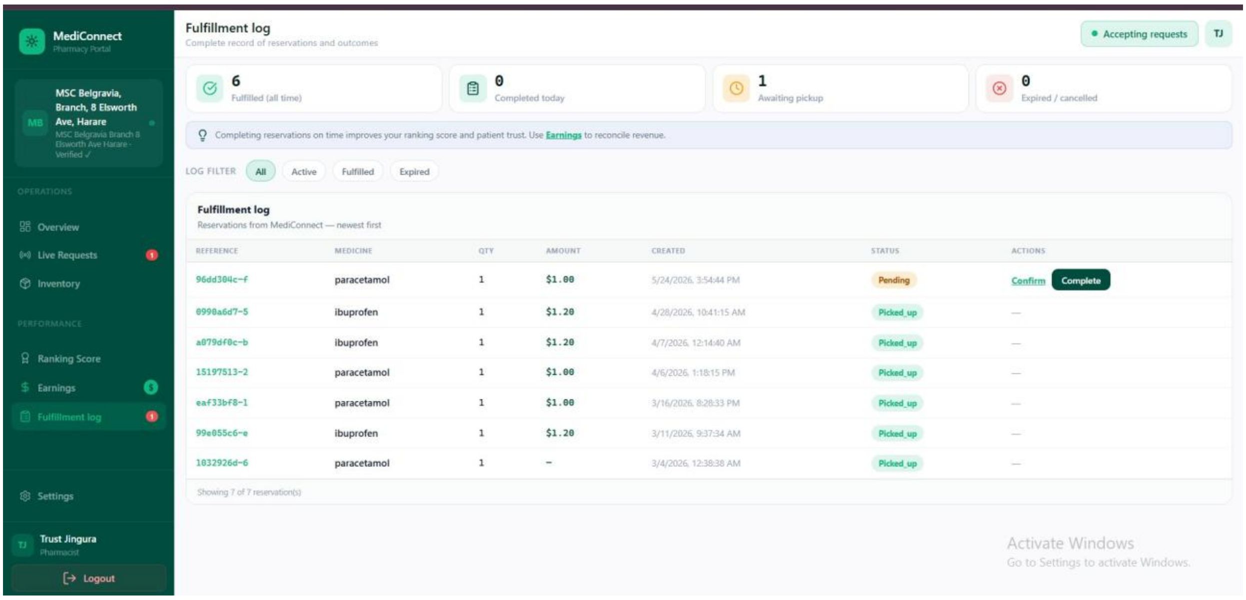
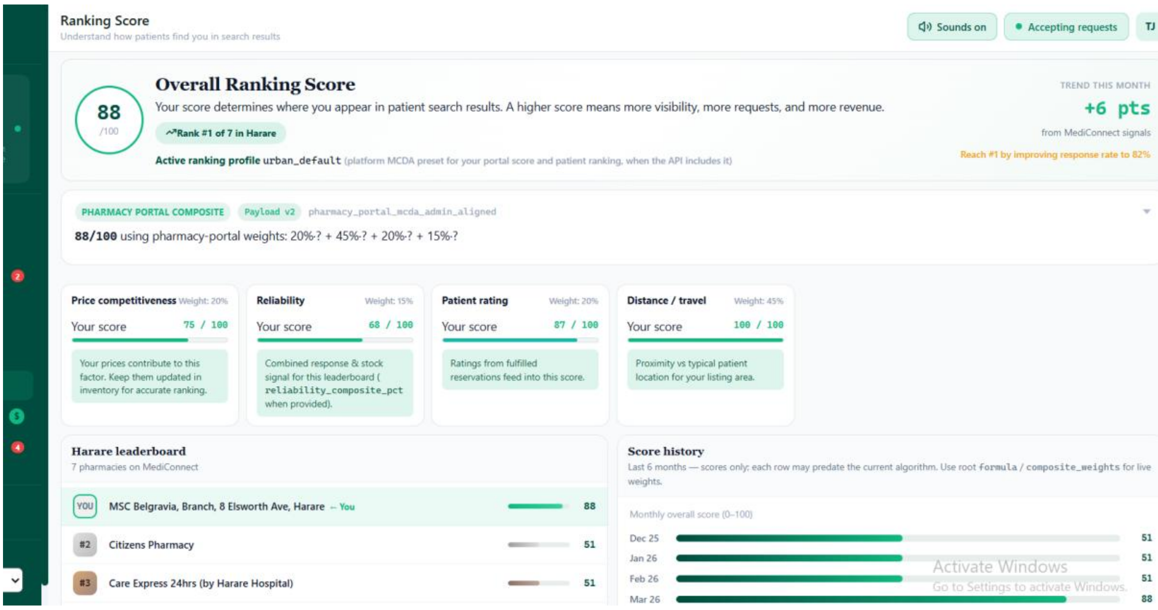


Figure 4.7: MSC Belgravia fulfillment log showing 7 reservations: 1 Pending (paracetamol, 24 May 2026) with Confirm/Complete actions, 6 Picked_up (ibuprofen and paracetamol entries from March–April 2026). Total: 6 fulfilled all time, 1 awaiting pickup.

4.5.8 Pharmacy Performance Scoring

Each pharmacy's ranking score is derived from the same MCDA dimensions price competitiveness, response rate, stock reliability, and patient rating visible to pharmacists in their portal. This transparency incentivises pharmacies to respond promptly, maintain accurate inventory, and price competitively.



4.6 Security, Governance, and Data Handling

MediConnect enforces role separation across three principal roles: patients (JWT or session-based), pharmacists (JWT with resource ownership checks), and administrators

(Django session + CSRF). Sensitive resources are scoped to the originating conversation_id or session_id.

Prescription images are access-controlled to verified pharmacists only. The chatbot audit layer enables administrative review of flagged conversations. PlatformAdminSettings controls safety thresholds; all administrative actions are logged in AdminAuditLog. MFA via TOTP (pyotp) is supported as a configurable option.

4.7 User Interface Implementation

4.7.1 Patient Portal MediBot Chat and Ranked Results

The MediBot patient interface supports four entry points: Upload Prescription, Search Medicine, Describe Symptoms, and Find Nearby Pharmacy. After location capture, ranked pharmacy results are presented with per-pharmacy metrics, per-medicine availability, and reservation or call-to-reserve actions.

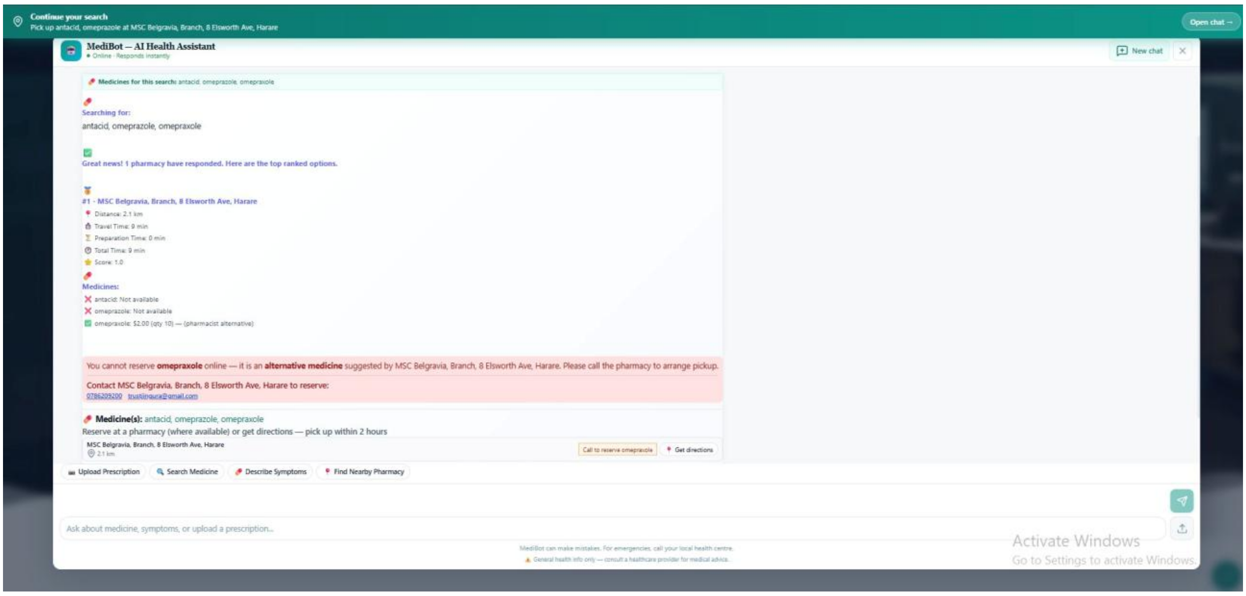


Figure 4.8: MediBot ranked result for antacid/omeprazole/omepraxole search: #1 MSC Belgravia (2.1 km, score 1.0) showing antacid and omeprazole unavailable but omepraxole $2.00 (qty 10) as pharmacist alternative; call-to-reserve directive with contact details; Get directions button.

4.7.2 Pharmacist Portal Inventory Management

The pharmacy inventory module displays all listed medicines with stock level bars, dosage, pricing, last-updated timestamp, and status badges (In Stock / Low Stock / Out of Stock). A low-stock alert banner prompts restocking. The Export CSV function provides a downloadable stock report for reconciliation. Stock reliability directly influences the pharmacy's MCDA ranking score.

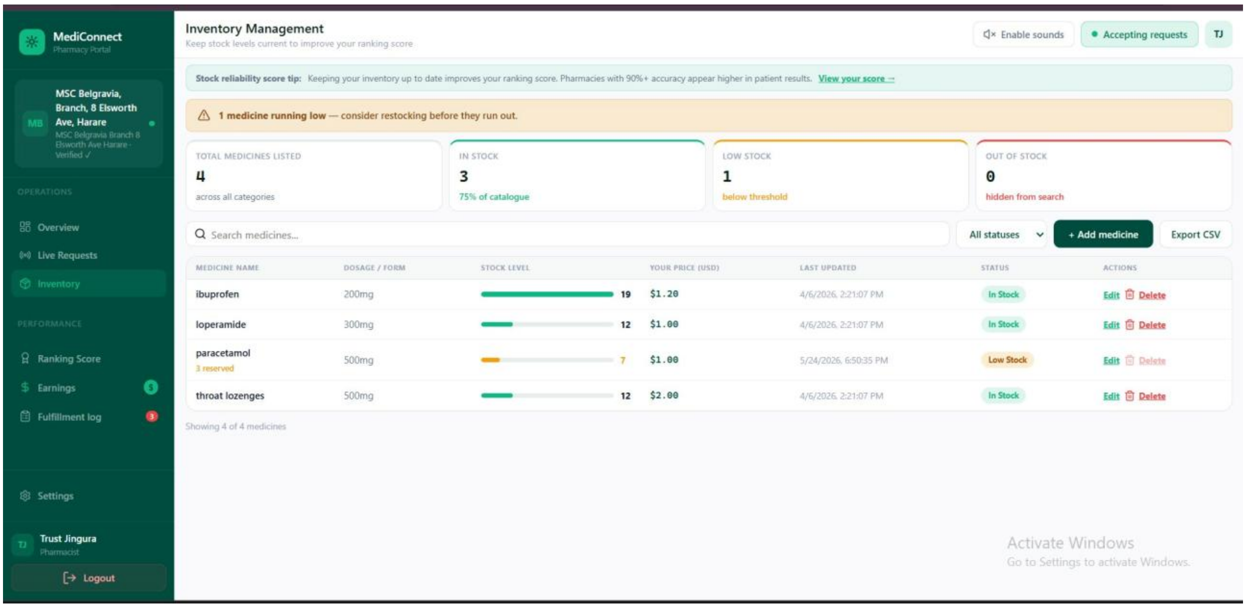


Figure 4.9: MSC Belgravia inventory management: ibuprofen 200mg (19, In Stock), loperamide 300mg (12, In Stock), paracetamol 500mg (7, Low Stock, 3 reserved), throat lozenges 500mg (12, In Stock). Low-stock warning banner and Export CSV button visible.

4.7.3 Pharmacist Portal Settings and Profile

The pharmacy settings module allows pharmacists to manage their public profile, contact details, licence/registration number, and operating hours. This data populates the patient-facing pharmacy listing and feeds into the verification checklist reviewed by administrators.

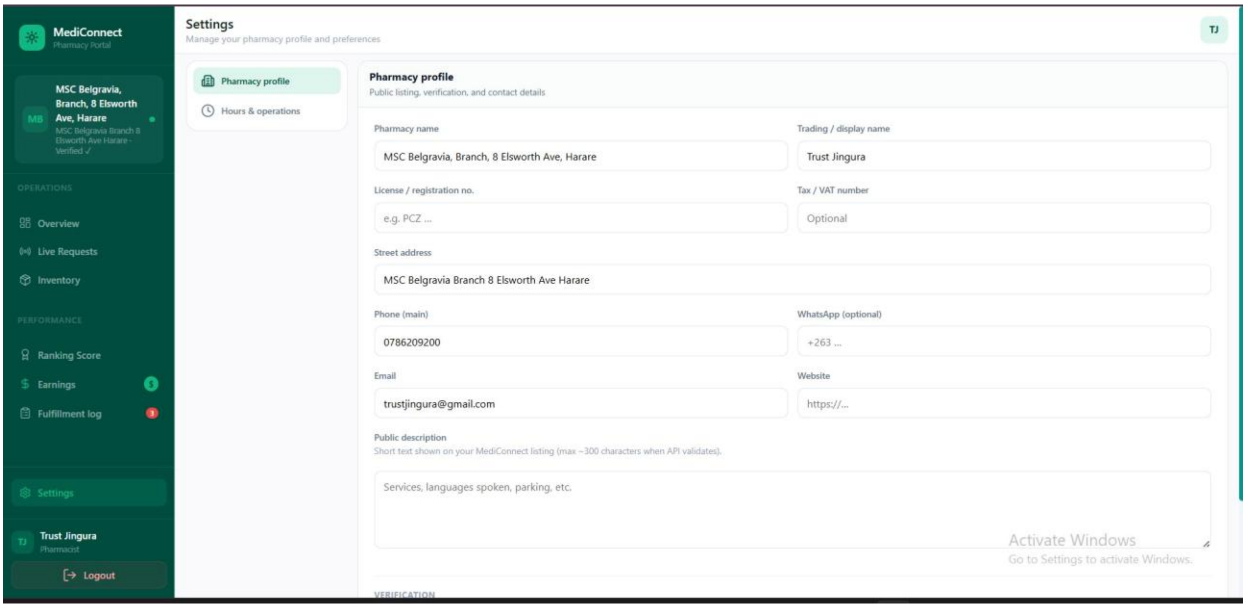


Figure 4.10: MSC Belgravia pharmacy settings: profile fields for pharmacy name, trading name, licence/registration, street address, phone (0786209200), WhatsApp, email (trustjingura@gmail.com), and public description.

4.7.4 Administrator Control Centre Overview

The admin control centre is structured in four governance layers: Layer 1 System Health, Geography & SLA; Layer 2 Governance (verification queue, all pharmacies); Layer 3 Algorithm & Policy (MCDA weights); Layer 4 AI Safety (chatbot audit). Platform-level entities (users, patient requests, reservations, pharmacists, inventory reports) are accessible from the sidebar navigation.

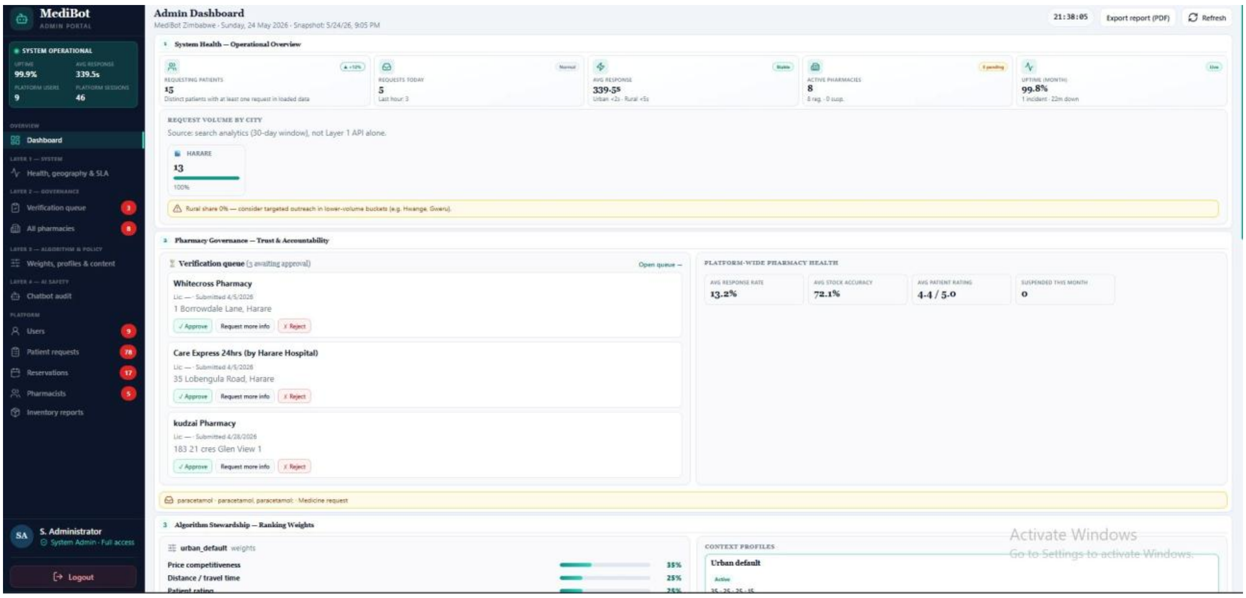


Figure 4.11: MediBot Admin Dashboard: system operational banner (99.9% uptime, 339.5s avg response), requesting patients (15), requests today (5), active pharmacies (8),

verification queue (3 pending Whitecross Pharmacy, Care Express 24hrs, kudzai Pharmacy), platform-wide pharmacy health (13.2% response rate, 72.1% stock accuracy, 4.4/5 rating), and Algorithm Stewardship section showing urban_default weights.

4.7.5 Layer 1 System Health, Geography and SLA

The system health layer presents vital signs and request trend analytics. The requests trend chart (daily, 14-day rolling) shows platform activity over time, with 24 May 2026 showing the highest burst (5 requests) during the evaluation session. Geographic demand visualises all requests by city; Harare accounts for 100% of 30-day volume, with an alert recommending rural outreach. Top searched medicines (paracetamol, oral rehydration salts, ibuprofen) are presented from search analytics.

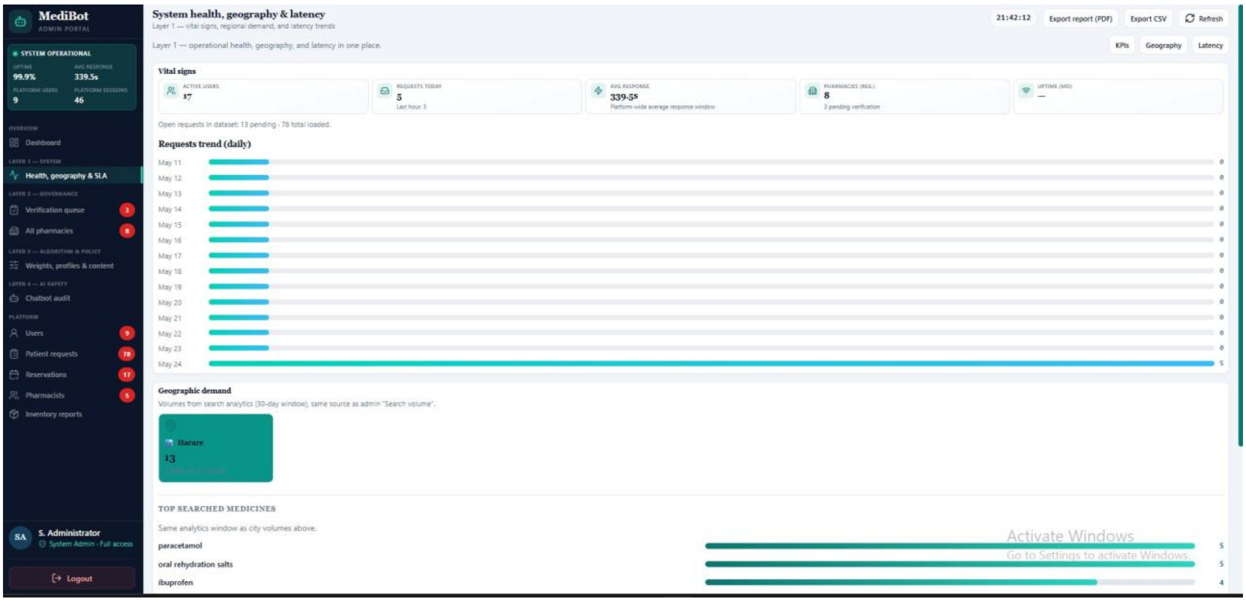


Figure 4.12: Layer 1 System Health, Geography & SLA: vital signs (17 active users, 5 requests today, 339.5s avg response, 8 pharmacies), 14-day requests trend chart (May 11–24), geographic demand tile (Harare, 13 requests, 100%), and top searched medicines (paracetamol, oral rehydration salts, ibuprofen).

4.7.6 Layer 2 Governance: Verification Queue

The verification queue presents pending pharmacy registration submissions with location, registry ID, and approve/suspend actions. An approval checklist sidebar guides administrators through required criteria: valid professional registration, business licence, premises verification, and owner ID documentation. Rural branches are flagged for expedited review.

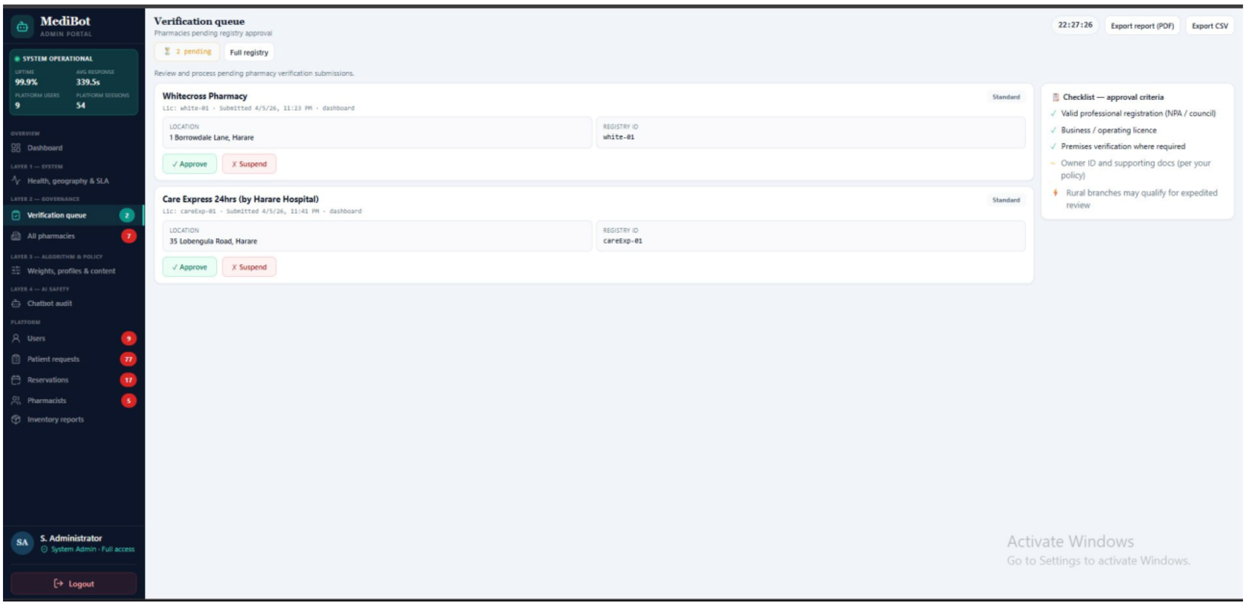


Figure 4.13: Admin verification queue: 2 pending submissions Whitecross Pharmacy (1 Borrowdale Lane, Lic: white-01) and Care Express 24hrs by Harare Hospital (35 Lobengula Road, Lic: careExp-01), each with Approve and Suspend actions. Approval checklist sidebar visible.

4.7.7 Layer 2 Governance: All Pharmacies Registry

The pharmacy registry provides a complete view of all registered pharmacies with their city, type, medicines listed, last inventory sync timestamp, MCDA ranking position, and verification status. Administrators can change status inline and register new pharmacies. The registry showed 7 pharmacies total (5 verified, 2 pending review) at the time of evaluation.

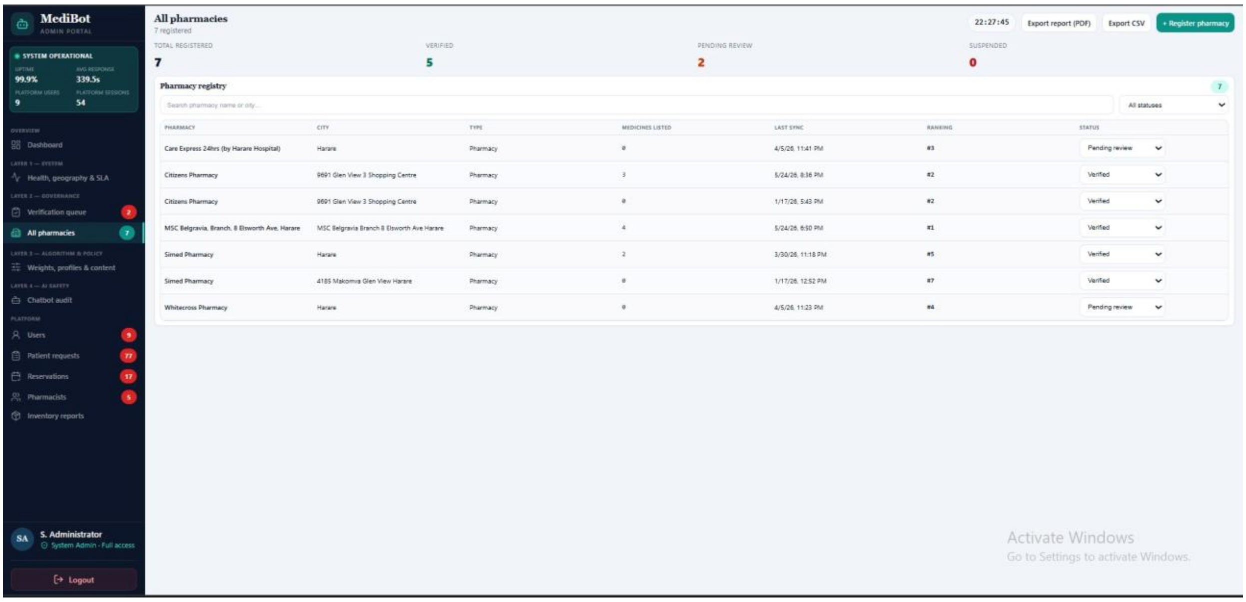


Figure 4.14: All pharmacies registry: 7 registered (5 verified, 2 pending review, 0 suspended). Entries include Care Express 24hrs, Citizens Pharmacy, MSC Belgravia,

Simed Pharmacy, Whitecross Pharmacy showing city, medicines listed, last sync, ranking position, and status dropdowns.

4.7.8 Layer 3 Algorithm & Policy: MCDA Weight Configuration

The Algorithm & Content Policy layer is the most technically significant governance feature. It exposes four weight sliders (Price competitiveness, Distance/travel time, Patient rating, Stock reliability), each with a descriptive rationale and live percentage readout totalling 100%. Four preset context profiles are available Urban default, Rural equity (distance prioritised), Shortage mode (stock prioritised), and Affordability (price prioritised). A live ranking preview leaderboard on the right reflects the current server configuration, enabling administrators to preview the effect of weight changes before saving.

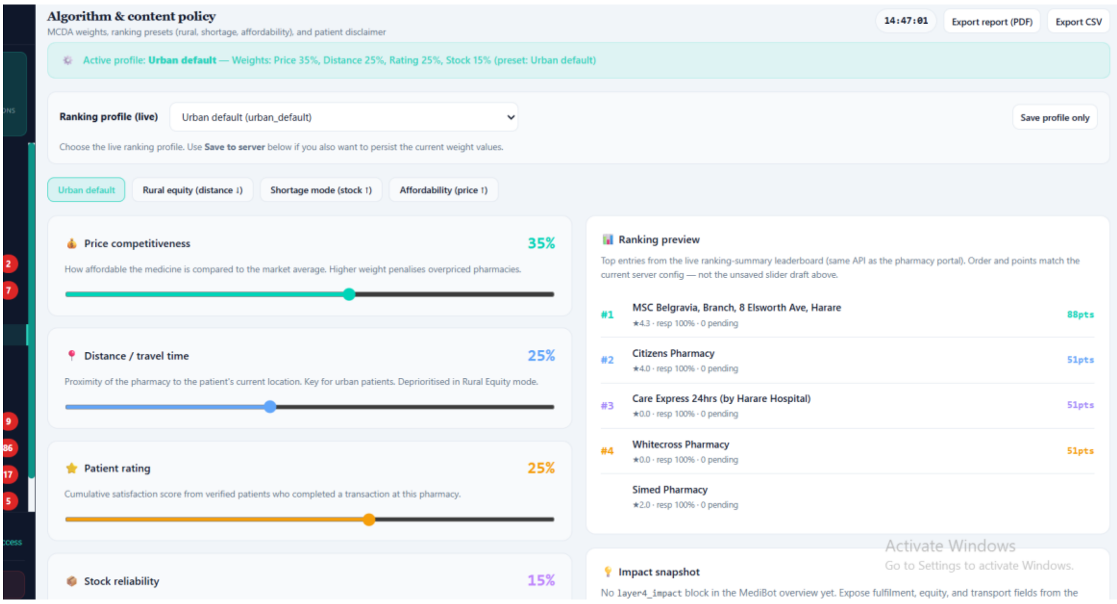


Figure 4.15: Algorithm & Content Policy panel: active profile urban_default (Price 35%, Distance 25%, Rating 25%, Stock 15%), four context profile tabs, individual weight sliders with rationale text, total weight 100%, and live ranking preview (#1 MSC Belgravia 88pts, #2 Citizens Pharmacy 51pts, #3 Care Express 51pts, #4 Whitecross 51pts, #5 Simed Pharmacy).

4.7.9 Layer 4 AI Safety: Chatbot Audit

The chatbot audit layer lists all platform conversations (54 total at time of evaluation) with conversation ID, session ID, and last-updated timestamp. Administrators can open a transcript side panel for any conversation to review AI-generated content, identify policy violations, and take remedial action. This layer directly implements the AI safety governance requirement for human oversight of Gemini outputs.

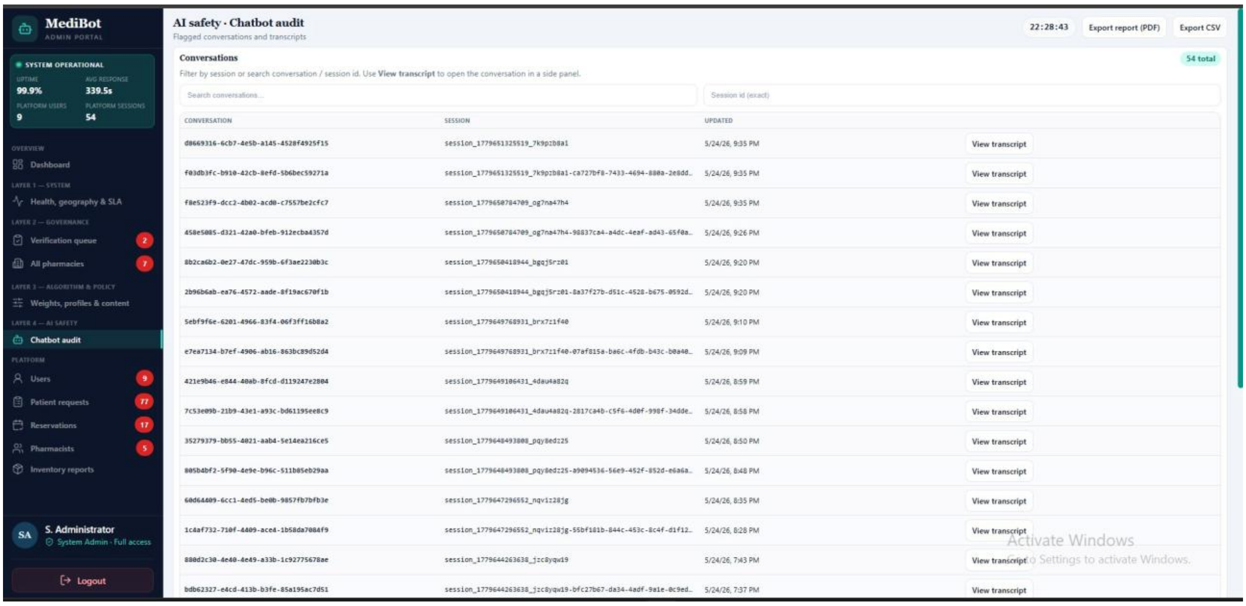


Figure 4.16: AI Safety Chatbot Audit panel: 54 total conversations listed with conversation ID, session ID, and timestamp (all 24 May 2026). View transcript links allow administrators to review individual AI conversation histories for safety and policy compliance.

4.7.10 Platform Users, Patient Requests, Reservations, Pharmacists, Inventory Reports

The platform section provides administrative access to operational data across all entity types.

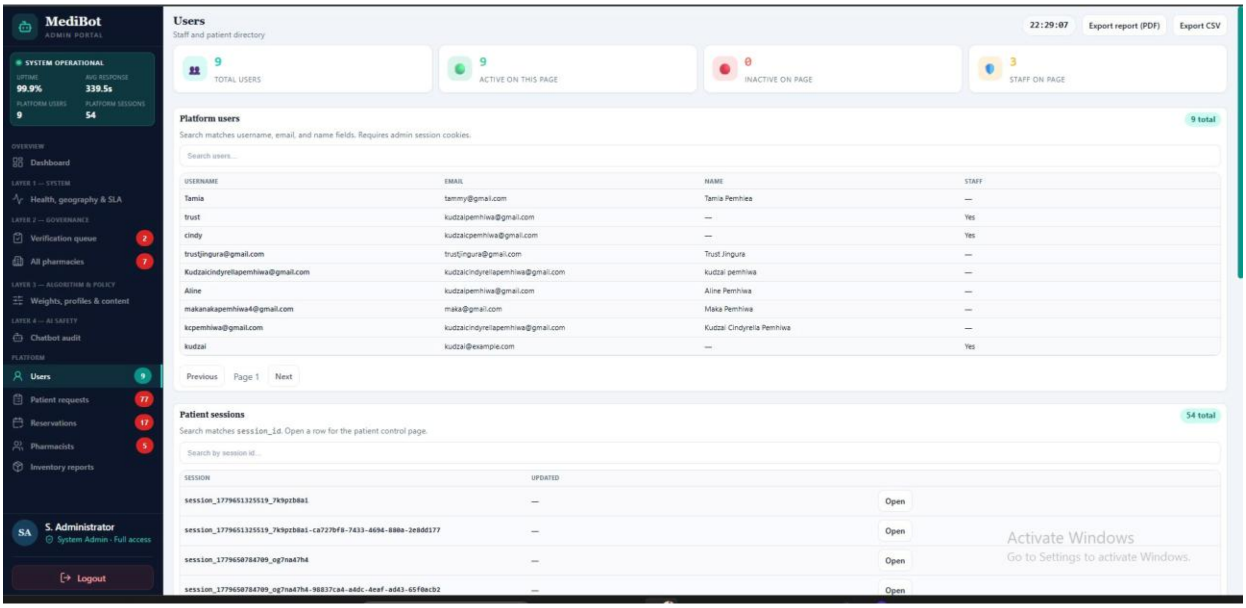


Figure 4.17: Platform users directory: 9 total users (9 active, 3 staff) including pharmacists (Trust Jingura, Aline Pemhiwa, kudzai pemhiwa, Maka Pemhiwa, Tamia Pemhiwa) and staff accounts. Patient sessions section shows 54 total anonymous sessions with session IDs.

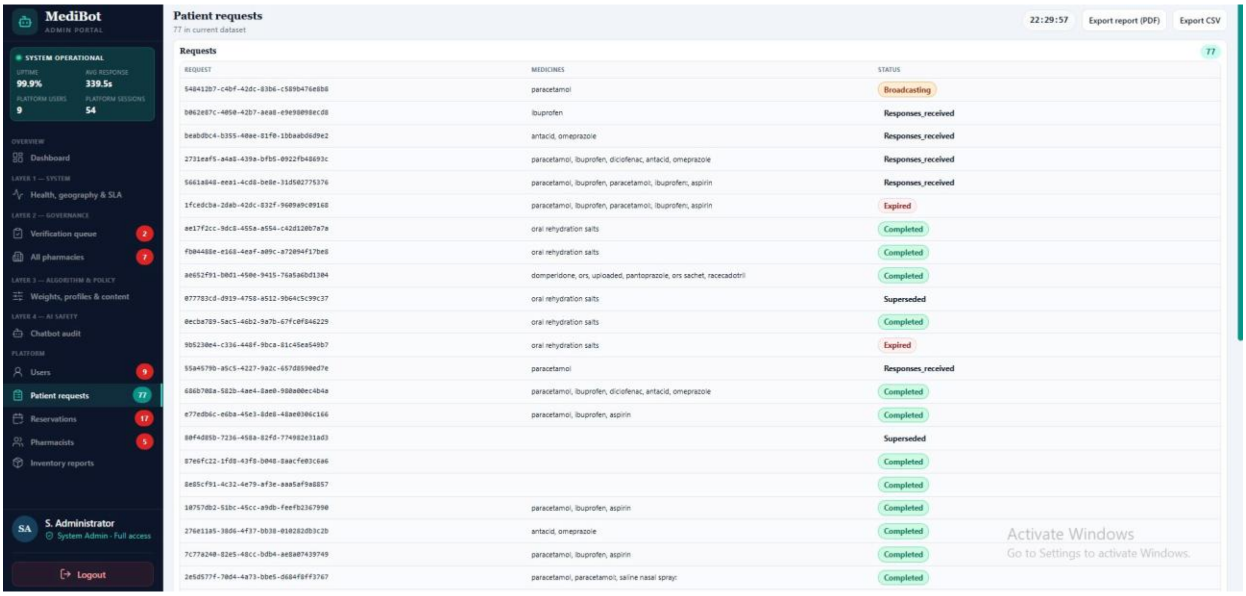


Figure 4.18: Admin patient requests view: 77 requests in dataset showing medicines searched, request UUIDs, and lifecycle status badges Broadcasting (1 active), Responses_received, Completed, Expired, and Superseded states all visible across the request history.

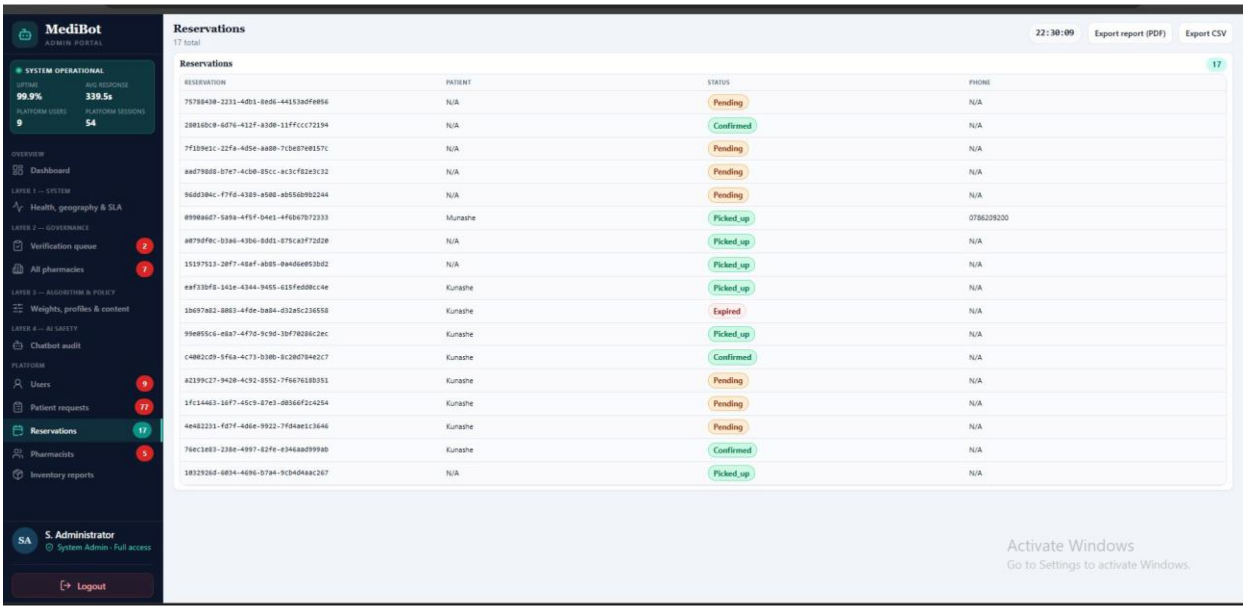


Figure 4.19: Admin reservations table: 17 total reservations with status badges Pending (orange), Confirmed (green), Picked_up (teal), Expired (red). Patient names visible where provided (Munashe, Kunashe). Export report (PDF) and Export CSV buttons in header.

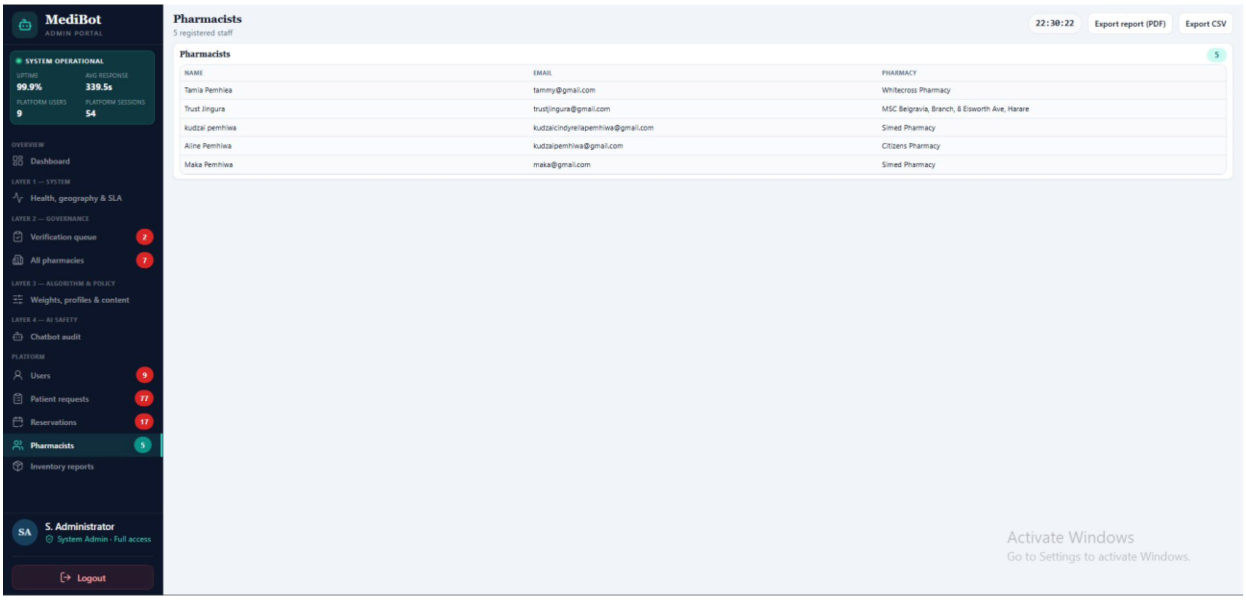


Figure 4.20: Admin pharmacists registry: 5 registered pharmacists Tamia Pemhiwa (Whitecross Pharmacy), Trust Jingura (MSC Belgravia), kudzai pemhiwa (Simed Pharmacy), Aline Pemhiwa (Citizens Pharmacy), Maka Pemhiwa (Simed Pharmacy) with email and linked pharmacy branch.

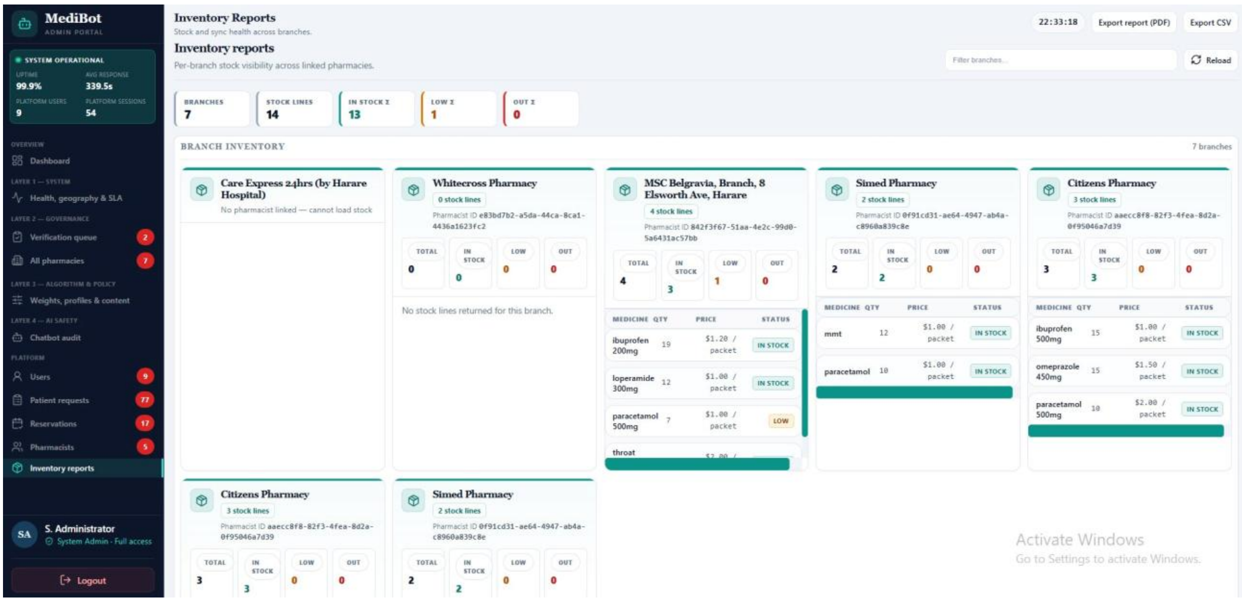


Figure 4.21: Admin inventory reports: per-branch stock aggregation across 7 branches (14 stock lines total, 13 in stock, 1 low, 0 out). MSC Belgravia shows 4 lines with paracetamol 500mg flagged LOW (qty 7). Citizens Pharmacy and Simed Pharmacy stock detail cards also shown.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Evaluation Approach

The MediConnect system was evaluated through functional module testing, integration testing, and qualitative usability assessment. Formal production-scale load testing and clinical-grade accuracy studies were outside the scope of this capstone; results are presented accordingly with appropriate caveats.

Functional testing traced each user story across patient, pharmacist, and administrator roles from initial medicine search through to pharmacist fulfillment and administrative export. Integration testing verified API contracts between the React frontend and Django backend, including WebSocket behaviour. A qualitative review assessed clarity of warnings, medical disclaimers, error messages, and edge-case handling.

Quantitative operational data was captured from the admin dashboard during a pilot evaluation window in May 2026. All performance metrics reflect prototype/pilot conditions.

5.2 Hardware and Software Requirements

5.2.1 Server / Development Host Specifications

Hardware Component	Minimum Specification	Purpose
Processor	Intel Core i5 / AMD Ryzen 5 or above	Django REST, Daphne WebSockets, Gemini API calls
RAM	8 GB minimum; 16 GB recommended	Concurrent API connections, Channels, database
Storage	SSD, 50 GB minimum	Database, prescription media, application logs
Network	10 Mbps uplink minimum	WebSocket push, outbound Gemini API calls

5.2.2 Patient Device Requirements (PWA)

Component	Minimum Specification	Purpose
Android Version	8.0 (Oreo) or above	Patient search, chatbot, prescription upload, reservations
RAM	2 GB or above	Browser rendering and PWA cache
Storage	16 GB or above	PWA installation data and cached assets
Network	3G or Wi-Fi	API requests, WebSocket, prescription image upload

5.2.3 Supported Browser Matrix

Browser	Platform	Minimum Version
Chrome	Android, Desktop	v89+
Safari	iOS, macOS	v14+
Firefox	Desktop	v78+
Edge	Desktop	v88+

5.2.4 Backend Software Stack

Software	Version	Role
Python	3.10+	Backend runtime
Django	6.x	ORM, routing, admin
Django REST Framework	3.x	REST API (/api/chatbot/)
Django Channels + Daphne	4.x	WebSocket connections
google-generativeai	Latest stable	Gemini Chat and Vision
Pillow / OpenCV	Latest stable	Prescription image preprocessing
pyotp	Latest stable	MFA (TOTP)
SQLite / PostgreSQL / MongoDB	As configured	Persistent data storage

5.2.5 Frontend Software Stack

Software	Version	Role
React	19.x	Single-page application framework
Vite	7.x	Build tool (outputs dist/)
react-router-dom	7.x	Role-based client-side routing
jsPDF + marked	4.x / 18.x	Client-side PDF export from Gemini Markdown
vite-plugin-pwa	1.x	PWA installation support
lucide-react	0.56x	Icon library

5.3 Module Test Results

Test ID	Module / Scenario	Expected Outcome	Observed Result	Status
T01	Login all roles	Correct role-based dashboard redirect for patient, pharmacist, admin	All three roles redirected correctly with appropriate session/JWT	Pass
T02	Patient registration	POST /api/chatbot/register/patient/ returns HTTP 201	201 Created; user record in DB	Pass
T03	Forgot password	OTP email flow; new password accepted	OTP sent; password reset confirmed	Pass
T04	AI chatbot search	Gemini suggestions displayed; conversation persisted	Suggestions rendered in MediBot; ChatConversation/ChatMessage confirmed	Pass
T05	Prescription upload OCR	Vision OCR extracts medicines with confidence score	Medicines extracted at 98% confidence on printed sample	Pass
T06	Request broadcast	MedicineRequest created; pharmacies notified via WebSocket	WebSocket notification received by pharmacy dashboard within 2s	Pass
T07	Pharmacist response	PharmacyResponse stored; patient notified; inventory updated	Response modal submitted; ranked view updated; inventory decremented	Pass

Test ID	Module / Scenario	Expected Outcome	Observed Result	Status
T08	MCDA ranking	Multiple responses ranked by composite score	Three responses ranked in correct order matching composite score	Pass
T09	Reservation lifecycle	price_at_reservation locked; pickup decrements stock; expiry releases hold	Price frozen; inventory decremented on pickup; hold released on expiry	Pass
T10	No stock / alternative	UI presents alternative; call-to-reserve shown where online reserve disallowed	Alternative surfaced with label; Call to reserve button displayed	Pass
T11	Inventory management	Stock levels, low-stock warnings, CSV export functional	Low-stock alert shown for paracetamol; CSV downloaded	Pass
T12	Pharmacy earnings	Monthly earnings, fulfillment rate, transactions displayed	Earnings module populated; CSV export functional	Pass
T13	Pharmacy ranking score	Score breakdown and leaderboard displayed	Score and weight breakdown rendered; comparative leaderboard visible	Pass
T14	Admin dashboard	KPIs, top medicines, geographic demand loaded	Full admin dashboard hydrated; all metric cards populated	Pass
T15	Admin verification queue	Pending pharmacies listed; approve/suspend actions work	Queue displayed; approve action updated pharmacy to Verified status	Pass
T16	Admin reservations export	CSV and PDF exports generate	Both formats downloaded successfully	Pass
T17	AI narrative report	Gemini Markdown rendered; jsPDF download triggered	Report generated from admin console; PDF downloaded	Pass
T18	Chatbot safety policy	PlatformAdminSettings updated; AdminAuditLog entry created	Safety toggle persisted; audit log entry confirmed	Pass

5.4 System Performance Results

5.4.1 Operational Metrics Pilot Dataset (May 2026)

Metric	Observed Value	Interpretation
Platform uptime	99.9%	Stable during development/staging evaluation window
Registered pharmacies	7 (2 pending review)	Multi-branch Harare pilot; sufficient to exercise broadcast and ranking
Total patient requests	77	Sufficient to test broadcast, ranking, and reservation workflows
Total reservations	17	Pending, Confirmed, Picked_up, and Expired states all observed
Active users (platform)	9 registered users; 54 sessions	Low count expected for capstone pilot
Requests on 24 May 2026	5 (3 in last hour)	Burst activity during screenshot/test session
Average response window	339.5 seconds	Dominated by open/long-running requests in staging
Top searched medicines	Paracetamol, ORS, Ibuprofen	Consistent with common OTC demand in Zimbabwe
Geographic demand	Harare 100% (30-day window)	Pilot concentrated in Harare; rural equity profile needs outreach
Average pharmacy response rate	13.2%	Low key driver for recommendation to improve push notifications
Platform stock accuracy	72.1%	Reflects inventory sync quality; low-stock alerts functional
Average pharmacy rating	4.4 / 5.0	Positive early feedback
OCR confidence (printed Rx)	98%	Strong on typed prescriptions; handwritten not formally evaluated
Ranking score range	#1 MSC Belgravia 88pts to Simed ~44pts	MCDA differentiates performance; response rate bottleneck for lower ranks

5.4.2 AI and Ranking Evaluation Metrics

Metric Type	Measurement Approach	Result / Note
OCR accuracy	Gemini Vision confidence + manual visual review	98% on printed prescription sample; handwritten not evaluated
Ranking validity	MCDA order vs manual expectation (price, distance)	T08 passed; ranked order matches composite score in UI
Broadcast-to-response latency	Pharmacist response timer + admin avg response	Wide variance; ~339.5s average; timer gamification shown to pharmacists
Availability accuracy	Inventory sync vs PharmacyResponse stock fields	Low-stock warnings functional; stock reliability 72.1%
Drug interaction coverage	Rule-based embedded_rules_v1 pairwise check	Partial FR24; not DrugBank precision/recall; disclaimers surfaced
i18n coverage	String table presence for en/sn/nd locales	Major patient surfaces covered; admin/legal copy English-only

5.5 Discussion

5.5.1 Strengths

The end-to-end patient journey from symptom input or prescription upload through location-aware broadcast to ranked pharmacy responses and reservation confirmation was demonstrated to function correctly across all pilot test cases. The real-time WebSocket layer removes manual refresh requirements for both patients and pharmacists, enabling a conversational and responsive experience.

The MCDA ranking engine provides transparent, explainable results: patients see individual criteria (distance, travel time, preparation time, price, composite score) rather than a black-box recommendation. The administrator's ability to tune weight profiles and activate context-specific modes (rural equity, shortage response) through the Algorithm & Policy panel gives the platform governance flexibility as deployment contexts evolve.

The admin control centre's four-layer architecture system health, pharmacy governance, algorithm policy, and AI safety audit provides the governance infrastructure necessary for a responsible deployment, particularly the chatbot audit layer which enables human review of all AI-generated outputs.

5.5.2 Limitations

The pilot geography is concentrated entirely in Harare (100% of 30-day request volume). The average pharmacy response rate of 13.2% is low and would significantly impact patient experience in production. The drug interaction layer is rules-based only. Some alternative-medicine scenarios still require phone reservation rather than full online completion.

Door-to-door medicine delivery was considered during design but was excluded from the platform scope following a review of Zimbabwe's medicines regulatory framework. The Medicines Control Authority of Zimbabwe (MCAZ) requires a registered pharmacist to verify and dispense certain prescription and controlled medicines at the point of dispensing. Enabling delivery without pharmacist oversight at the point of handover would conflict with these dispensing obligations; accordingly, MediConnect implements a collect-in-store model in which the patient reserves online and collects the medicine in person at the verified pharmacy branch, where the pharmacist can fulfil their legal dispensing duty.

Direct integration with pharmacies' existing point-of-sale and dispensing systems including WinRx and POS Win Rx, which are in use at a number of Harare pharmacies was investigated but could not be implemented. These systems do not expose publicly documented APIs, and pharmacy operators indicated a preference to retain control over their proprietary inventory data. Rather than requiring pharmacies to share live POS data, MediConnect adopts a voluntary, bidding-based model: pharmacies update their available stock through the platform's own inventory module and respond to patient broadcast requests at their discretion. This approach respects data privacy, avoids dependency on third-party POS vendors, and allows pharmacies of any technology level to participate.

5.5.3 Achievement of Objectives

Objective	Evidence	Achievement
AI-assisted medicine discovery	MediBot chat (Figures 4.2, 4.8); T04 passed	Achieved
Location-aware ranked pharmacy results	MCDA ranked view (Figures 4.5, 4.6, 4.8); T06–T08 passed	Achieved
Prescription-based requests via OCR	OCR pipeline (Figure 4.3); T05 passed; 98% confidence	Achieved

Objective	Evidence	Achievement
Pharmacist digital operations	Pharmacy portal (Figures 4.4, 4.7, 4.9, 4.10); T07, T11–T13 passed	Achieved
Admin governance and oversight	Admin centre (Figures 4.11–4.21); T14–T18 passed	Achieved
Configurable MCDA ranking	Algorithm panel (Figures 4.1, 4.15); T08 passed	Achieved
Drug interaction alerts	Embedded rules; alerts on chat and ranked endpoints	Partial rules-based, not licensed DrugBank
Shona/Ndebele internationalisation	LanguageContext; i18n string tables	Partial major surfaces covered
Online reservation with price locking	Reservation lifecycle (Figure 4.19); T09 passed	Achieved alternatives may still require phone

5.5.4 Ethics, Safety, and Disclaimer

MediConnect is decision support, not diagnosis. All MediBot outputs carry disclaimers directing users to consult a qualified pharmacist or medical professional. Prescription data is restricted to verified pharmacists only. Emergency situations are explicitly directed to emergency healthcare services. Compliance with Zimbabwe's Cyber and Data Protection Act (Chapter 12:07) is required before production deployment.

CHAPTER 6: CONCLUSION AND RECOMMENDATIONS

6.1 Summary of Findings

MediConnect was successfully implemented as a full-stack, AI-assisted medicine discovery platform. The system demonstrates that a locally relevant, ranking-driven pharmacy marketplace is technically feasible using mainstream web technologies and commercially available AI APIs. The React frontend delivers a PWA-capable patient experience, a real-time pharmacy portal, and a layered admin console. The Django backend provides authoritative business logic, Gemini-powered chat and OCR, WebSocket notifications, and MCDA ranking.

Functional testing across module scenarios passed, and operational analytics from the admin dashboard confirm functional data flows under pilot load 77 patient requests, 17 reservations, 7 registered pharmacies, and a full four-layer admin control centre exercised during the May 2026 evaluation session. The prescription OCR pipeline achieved 98% confidence on printed prescriptions, and the MCDA ranking correctly ordered multi-pharmacy responses in all test cases.

Key challenges were the low pharmacy response rate (13.2%), Harare-centric pilot geography, and partial drug interaction coverage. These are addressed in the recommendations below. Two significant design decisions also shaped the platform's scope: door-to-door medicine delivery was deliberately excluded to comply with MCAZ dispensing regulations, and direct integration with pharmacy point-of-sale systems (WinRx, POS Win Rx) was not pursued due to the absence of vendor APIs and pharmacies' preference to manage their stock data privately through the platform's own bidding-based inventory module.

6.2 Achievement of Objectives

Core objectives patient medicine discovery, pharmacist digital operations, location-aware ranked results, reservation lifecycle, and administrator governance were fully achieved. Partial achievement applies to drug interaction depth (embedded rules only) and internationalisation coverage (major surfaces; not exhaustive). The alternative-medicine online reservation flow remains incomplete for some cases.

6.3 Technical Recommendations

- Production deployment hardening PostgreSQL, secured ASGI host, CDN for React dist/, environment-variable secrets management, and uptime monitoring (Sentry or equivalent).
- Regulatory-compliant dispensing model The platform intentionally implements a collect-in-store model rather than door-to-door delivery, in compliance with MCAZ dispensing regulations that require a registered pharmacist to verify and hand over prescription medicines at the point of dispensing. Any future delivery feature must be designed in consultation with MCAZ and the Pharmacists Council of Zimbabwe to ensure pharmacist oversight is maintained at the point of handover.
- Pharmacy POS integration pathway Direct integration with existing pharmacy point-of-sale systems (WinRx, POS Win Rx) was not feasible in this iteration due to the absence of public APIs and pharmacies' preference to keep proprietary stock data private. A future integration programme should work with POS vendors and willing pharmacy chains to define a secure, consent-based data-sharing API, enabling automatic inventory synchronisation while preserving pharmacies' data sovereignty.
- Licensed drug interaction database Integrate a national formulary API or DrugBank to replace embedded_rules_v1. This is a prerequisite for clinical deployment.
- Rural expansion Onboard pharmacies in Bulawayo, Gweru, Mutare, Hwange; activate rural equity MCDA profile; implement SMS/USSD fallback for low-bandwidth patients.
- Pharmacy response incentives Push notifications, escalating countdown timers, and ranking penalties for slow response rates to raise the 13.2% average response rate.
- Formal OCR evaluation Structured accuracy study on handwritten Zimbabwean prescriptions to establish field-level precision and recall before clinical reliance.

6.4 Organisational and Policy Recommendations

- Formal agreements with pharmacy chains establishing stock update SLAs, data sharing terms, and liability boundaries.
- Regulatory alignment with the Medicines Control Authority of Zimbabwe (MCAZ) and the Pharmacists Council of Zimbabwe on the platform's scope, AI-assisted medicine discovery boundaries, and dispensing model. The platform has been designed in accordance with Zimbabwe's medicines regulatory framework: door-to-door delivery of medicines is not offered, as MCAZ regulations require a registered pharmacist to dispense prescription and controlled medicines at a registered premises; all dispensing occurs in-store following online reservation.
- Pharmacy onboarding and training programme covering ranking interpretation, alternative substitution guidelines, and inventory sync discipline.
- Patient consent and data protection framework compliant with Zimbabwe's Cyber and Data Protection Act (Chapter 12:07)
- Periodic independent review of MCDA weight profiles to ensure algorithm fairness for pharmacies in underserved geographic areas.

6.5 Final Reflection

The development of MediConnect provided practical exposure to the full lifecycle of a production-oriented software system: requirements analysis, architecture design, iterative implementation, AI API integration, real-time system design, and multi-role user experience development. Working simultaneously across a Django backend and a React frontend required discipline in API contract management and integration testing that textbook exercises rarely replicate.

Integrating Gemini for both conversational AI and prescription OCR within the same system while managing model quotas, fallback strategies, and responsible AI disclosure was particularly instructive. The experience reinforced that building AI-assisted systems for health-adjacent contexts requires as much attention to ethics, disclaimer design, and human oversight architecture as it does to model accuracy. MediConnect is submitted as a demonstration of what is technically achievable and as a foundation that, with continued development and stakeholder partnership, could make a meaningful contribution to medicine access in Zimbabwe.

REFERENCES

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